WEB ENGINEERING

PROJECT REPORT_ BCS(5A)

MULTILINGUAL MEDICAL REPORT

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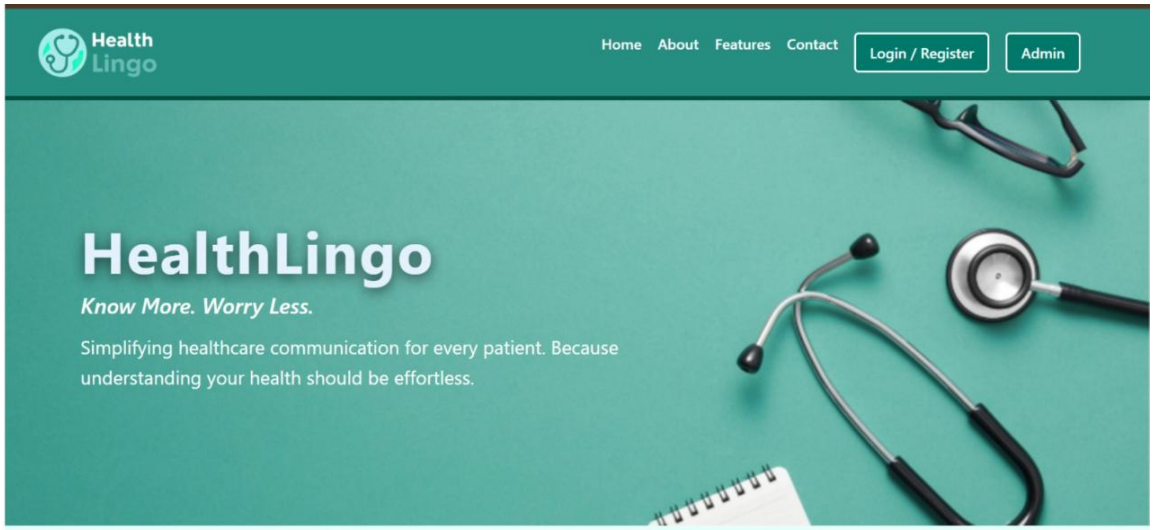
1.Introduction:

Medical reports are usually written for healthcare professionals and contain complex medical terminology, abbreviations, lab values, and clinical interpretations. Most patients, especially those with limited medical knowledge or low literacy levels, struggle to understand these reports. This lack of understanding can lead to anxiety, incorrect assumptions, delayed treatment, and poor health decisions.

The **Medical Report Simplification & Visualization System** is a web-based healthcare platform designed to bridge the communication gap between doctors and patients. The system uses **OCR, Natural Language Processing (NLP), rule-based severity analysis, visualization techniques, and multilingual support** to transform complex medical reports into **simple, understandable, and visual explanations**.

The system is especially beneficial for:

- Elderly patients
- Rural populations
- Low-literacy users
- Non-English speakers



2. Problem Statement:

Patients receive medical reports containing:

- Technical medical terms (e.g., Hemoglobin, Creatinine, Leukocytes)
- Numerical values without explanations
- No severity indication
- No visual reference to affected organs
- Language barriers

Due to this:

- Patients cannot understand whether their condition is serious or not.
- They rely on incorrect online sources.
- They panic unnecessarily or ignore serious issues.

There is **no unified platform** that:

- Simplifies reports
- Visualizes medical conditions
- Supports multiple languages
- Provides self-care guidance

3. Project Objectives:

The main objectives of this project are:

1. To simplify medical reports into patient-friendly language.
2. To provide severity classification (Normal, Mild, Moderate, Severe, Critical).
3. To visually represent affected organs using 2D/3D models.
4. To support multilingual output (English, Urdu, Punjabi).
5. To provide audio explanations using Text-to-Speech.
6. To allow secure sharing of reports.
7. To assist doctors in reviewing and providing feedback.
8. To improve patient awareness and healthcare decision-making.

4. Scope of the Project:

4.1 In-Scope

The project includes:

- Web-based user interface
- Secure login and registration
- Medical report upload
- OCR processing
- Medical term simplification
- Severity detection
- Visualization module
- Self-care instruction system
- Secure report sharing
- Doctor feedback module
- Admin management panel

4.2 Out of Scope

The following are not included:

- Emergency medical response
- Automatic diagnosis without doctor validation
- Online pharmacy
- Real-time wearable data

4.3 Benefits

- Patients understand medical reports easily.
- Doctors can efficiently manage and review patient data.
- Improves patient engagement and health literacy.

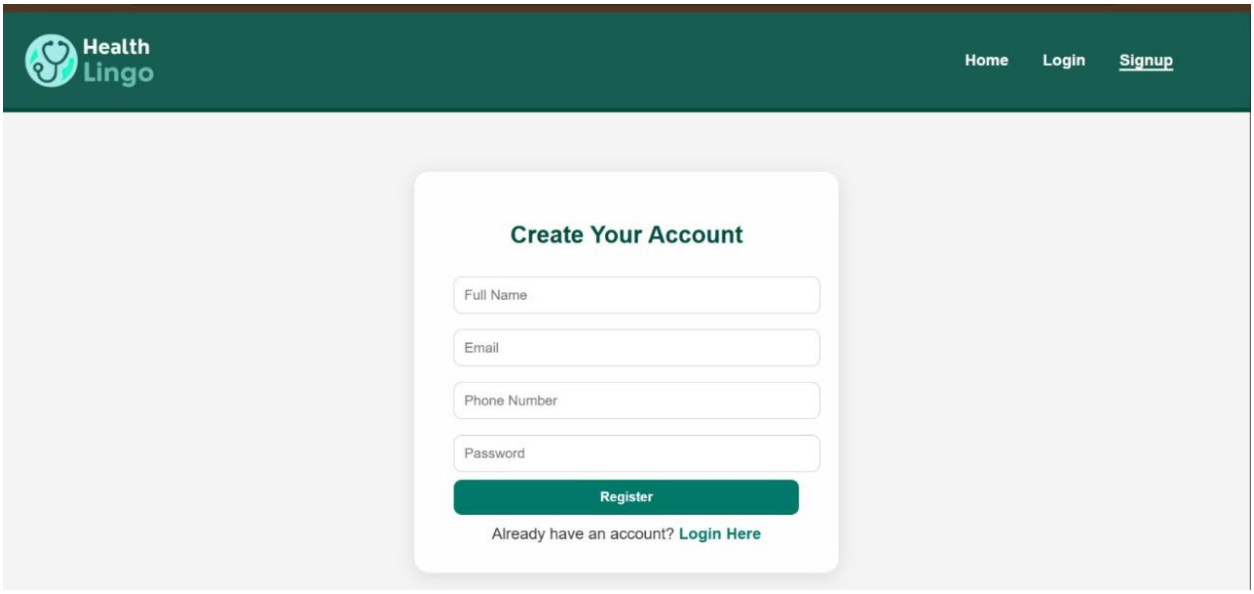
5. User Requirements:

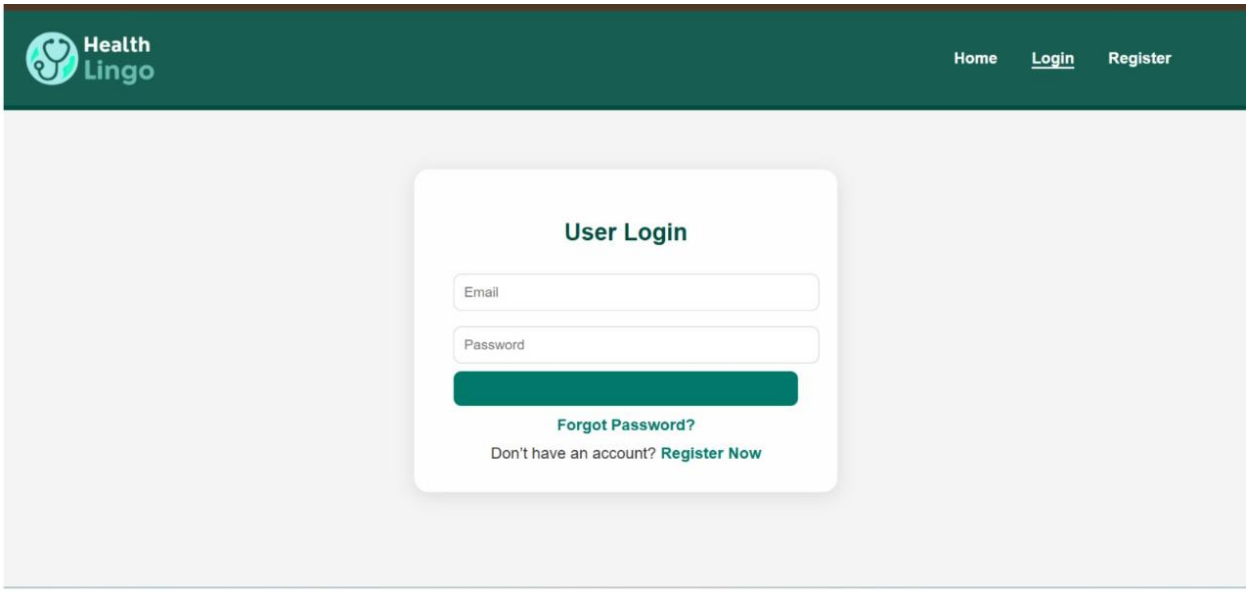
5.1 Patient Requirements

A patient should be able to:

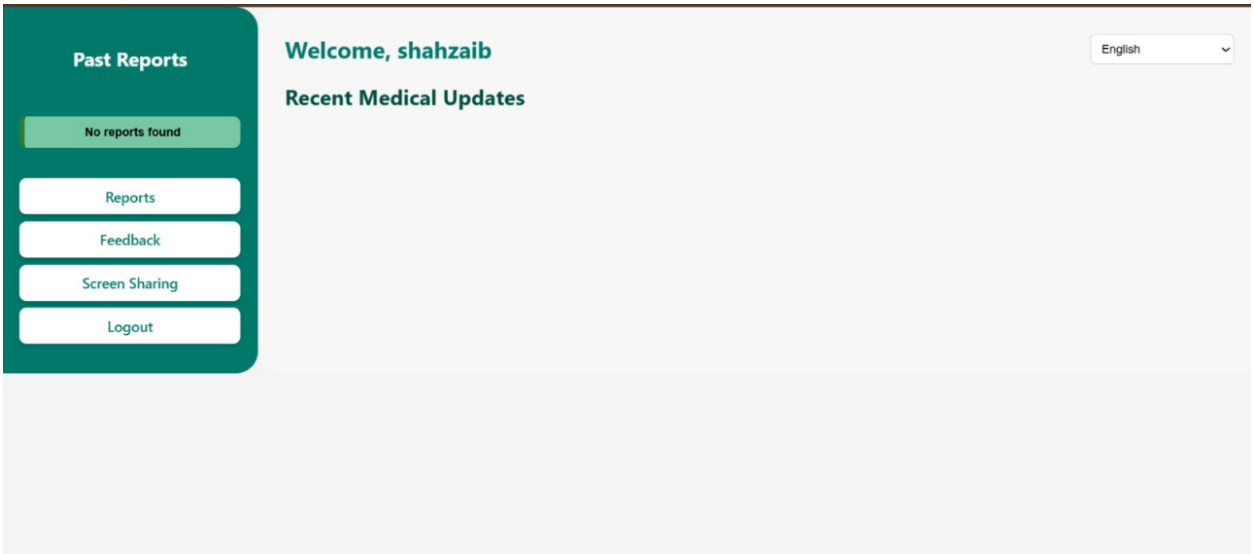
- Create an account using email or phone number.
- Log in securely using password.
- Upload medical reports in PDF, image, or text format.
- Capture reports using mobile camera.
- View simplified explanations.
- Understand severity using color codes.
- Translate reports into different languages.
- Listen to audio explanations.
- View past reports.
- Share reports securely.
- Receive alerts and notifications.

Patient Login Screen:

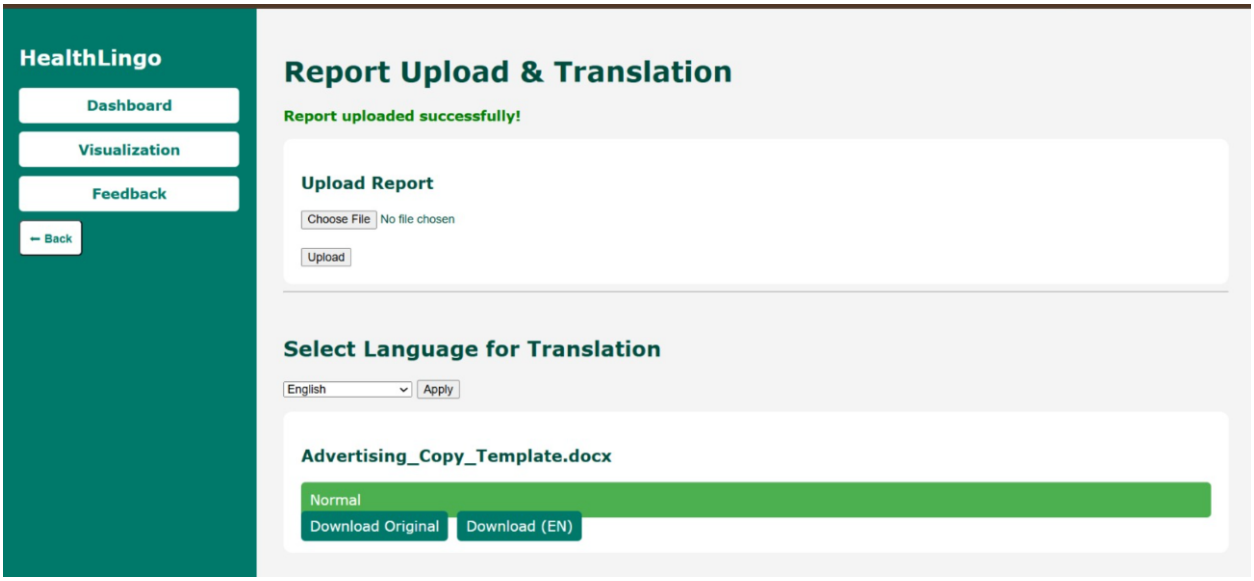




Patient Dashboard:



Upload Report Screen:

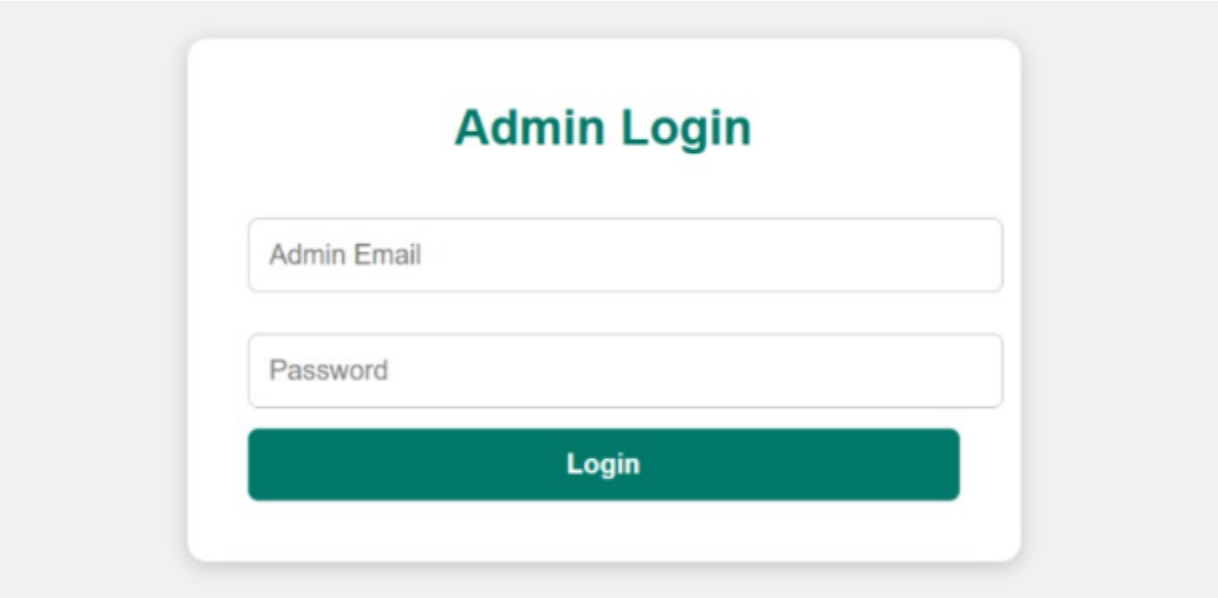


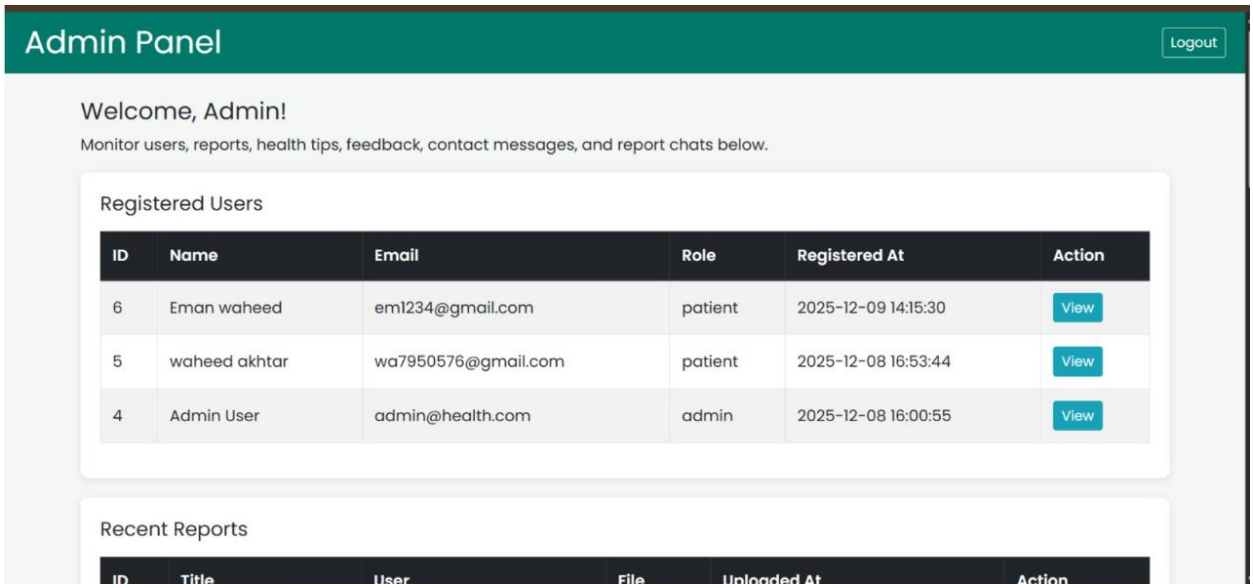
5.2 Admin Requirements

Admin should be able to:

- Manage users
- Monitor system performance
- Update medical content
- Maintain system security

Admin Panel Dashboard:





6. Functional Requirements:

Each system function is explained below:

- **User Authentication:** Secure login for patients and doctors.
- **Report Upload:** Supports PDF, image, text, and camera capture.
- **OCR Processing:** Extracts text from scanned reports.
- **Simplification Engine:** Converts medical terms into plain language.
- **Severity Analysis:** Uses thresholds and rules.
- **Translation Module:** Converts content into Urdu/Punjabi.
- **Text-to-Speech:** Converts text into voice.
- **Visualization Module:** Shows affected organs visually.

- **Sharing Module:** Secure sharing via links.
- **Profile Management:** Update user preferences.

7. Non-Functional Requirements:

- **Usability:** Simple UI for non-technical users.
- **Performance:** Fast processing within seconds.
- **Security:** Encrypted user data.
- **Reliability:** System uptime 24/7.
- **Scalability:** Supports future growth.
- **Accessibility:** Audio, font size, color contrast.

8. User Manual:

8.1 Patient Manual

Step 1: Register

- Navigate: **Signup → Patient Registration**
- Enter Name, Email/Phone, Password, optional medical history.
- Verify OTP sent to Email/Phone.
- *Screenshot Placeholder:* Patient Registration screen.

Step 2: Login

- Navigate: **Login → Patient Login**
- Enter Email/Phone + Password or OTP → Click **Login**
- *Screenshot Placeholder:* Patient Login screen.

Step 3: Upload Report

- Dashboard → **Upload Report** → Choose PDF/Image/Text or capture via camera
- OCR processes report automatically
- *Screenshot Placeholder:* Upload Report screen with progress bar

Step 4: View Simplified Report

- Dashboard → Report History → Select Report
- View **Original vs Simplified Report**, check **severity color codes**, Translate (English/Urdu/Punjabi), Text-to-Speech
- *Screenshot Placeholder:* Simplified Report screen

Step 5: Follow Self-Care Instructions

- Instructions displayed based on severity:

    - Normal/Mild → hydration, rest, diet
    - Moderate → suggested lifestyle changes
    - Severe/Critical → alert to consult admin/doctor immediately
- *Screenshot Placeholder:* Self-Care Instruction screen

Step 6: Share Report

- Click **Share**, generate secure link/QR code to share with family or authorized personnel
- *Screenshot Placeholder:* Share Report screen

8.2 Admin Manual

Step 1: Login

- Navigate: **Login → Admin Login**
- Enter Email + Password → Click **Login**
- *Screenshot Placeholder:* Admin Login screen

Step 2: Manage Users

- Dashboard → User Management
- View, Add, Edit, or Delete Patients
- *Screenshot Placeholder:* Admin User Management screen

Step 3: Monitor System

- Dashboard → System Status
- Check uptime, performance metrics, and alerts
- *Screenshot Placeholder:* System Monitoring screen

Step 4: Update Content

- Dashboard → Content Management
- Update infographics, self-care instructions, or other system content
- *Screenshot Placeholder:* Admin Content Update screen

9. Technical Manual:

9.1 System Architecture

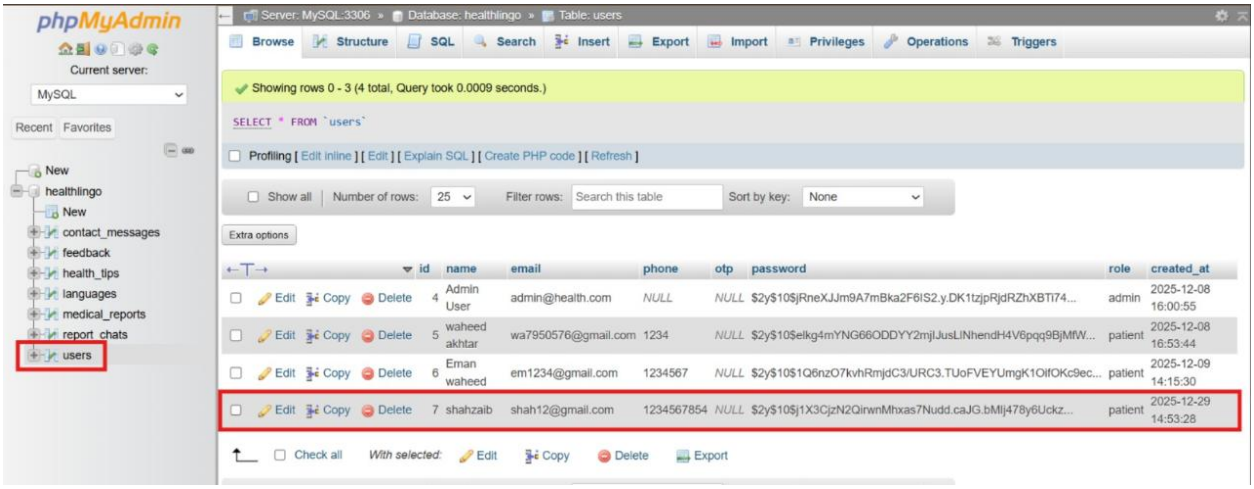
The system follows a layered architecture:

- Presentation Layer (Frontend)
- Application Layer (Backend APIs)
- Processing Layer (OCR, AI)

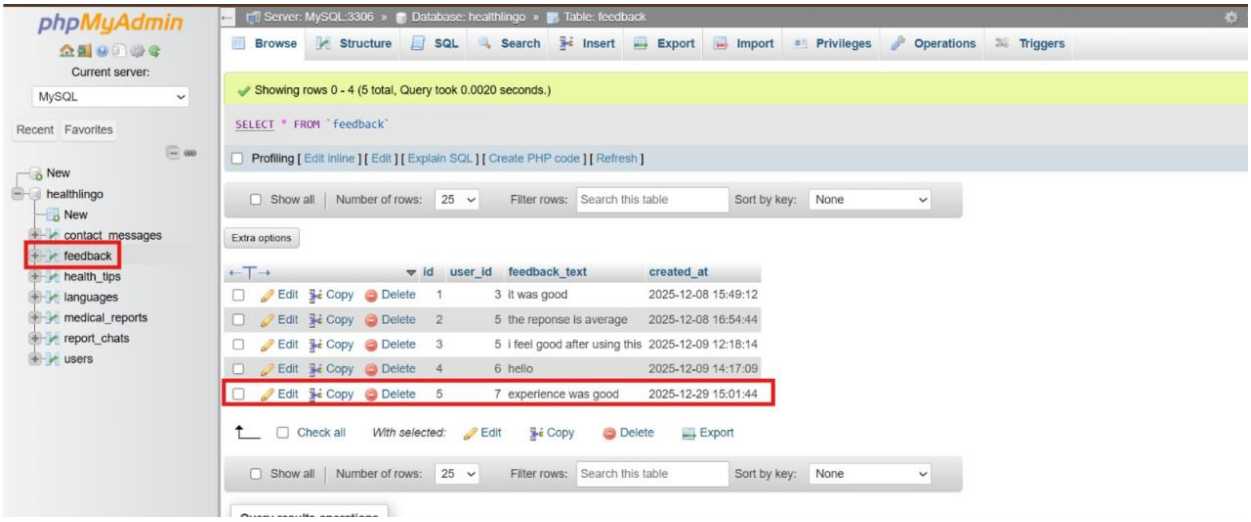
9.2 Technology Stack

Layer	Technology Used
Frontend	HTML, CSS, JavaScript
Backend	PHP (WAMP Server)
Database	MySQL
OCR	Tesseract OCR
Processing Logic	Rule-based logic (PHP)
Visualization	JavaScript (2D Charts / Icons)
Security	PHP Sessions, HTTPS
Server Environment	WAMP (Windows, Apache, MySQL, PHP)

User :



Feedback:



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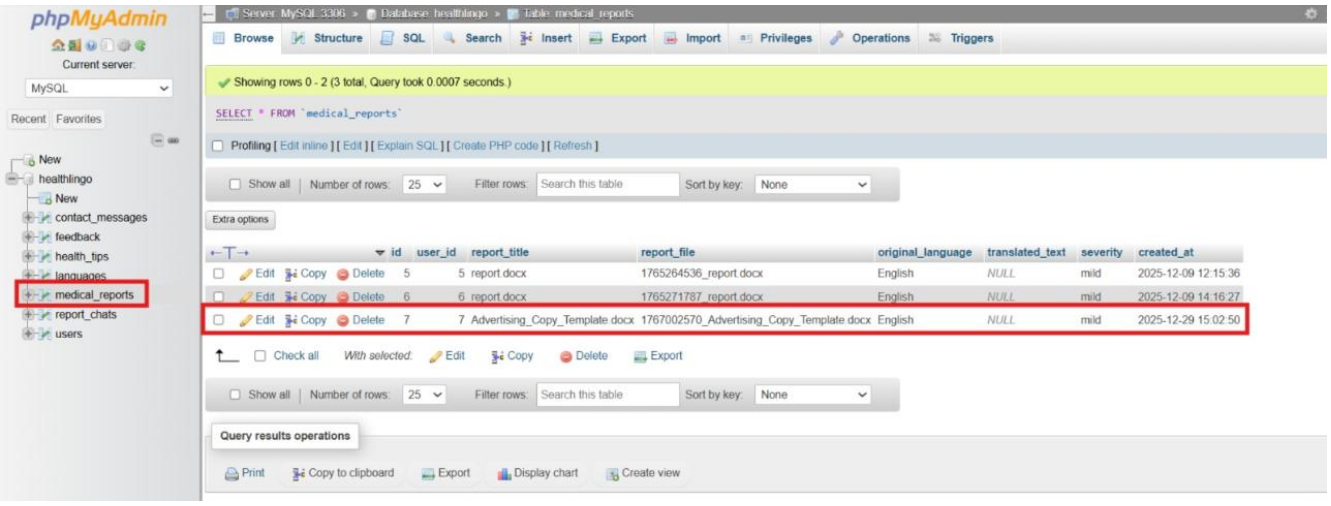

9.3 Security Measures

- Encrypted passwords
- Secure API tokens
- Role-based access
- Secure report sharing

10. Future Enhancements:

- Mobile app
- AI-based disease prediction
- Voice-based interaction
- Doctor video consultation
- Wearable device integration

11. Conclusion:

The **Medical Report Simplification & Visualization System** improves patient understanding, supports doctors, and promotes better healthcare communication. By combining AI, visualization, and multilingual accessibility, the system provides a **patient-centered healthcare solution**.